

COUNTEREXAMPLES TO THE $(v/2)$ -COMPRESSION CONJECTURE FOR PERIODIC GOLAY PAIRS

ABSTRACT. Lumsden, Kotsireas, and Bright conjectured that the $(v/2)$ -compression of every periodic Golay pair is equivalent to one of two forms determined by its row sums. We construct a binary Golay pair of length 100 whose 50-compression is

$$([12, 2], [-4, 6]),$$

and is equivalent to neither form. Since the conjecture was established for all $v < 100$, this is a least-length counterexample. Iterated Turyn multiplication gives counterexamples of length 10^n for every $n \geq 2$.

1. THE CONJECTURE AND AN OBSTRUCTION

For a sequence $X = (x_0, \dots, x_{v-1})$, write

$$N_X(s) = \sum_{j=0}^{v-1-s} x_j x_{j+s}, \quad P_X(s) = \sum_{j=0}^{v-1} x_j x_{j+s \bmod v}.$$

A pair (X, Y) of $\{\pm 1\}$ -sequences is a *Golay pair* if $N_X(s) + N_Y(s) = 0$ for $1 \leq s < v$, and a *periodic Golay pair* if $P_X(s) + P_Y(s) = 0$ for $1 \leq s < v$. Since

$$P_X(s) = N_X(s) + N_X(v-s),$$

every Golay pair is periodic.

For even v , let

$$r(X) = \sum_{j=0}^{v-1} x_j, \quad e(X) = \sum_{j=0}^{v-1} (-1)^j x_j.$$

The $(v/2)$ -compression of X is the length-two sequence

$$X^{(v/2)} = \left[\frac{r(X) + e(X)}{2}, \frac{r(X) - e(X)}{2} \right].$$

Lumsden, Kotsireas, and Bright conjectured that if the $(v/2)$ -compression of a periodic Golay pair has row sums a and b , then it is equivalent to one of

$$(1) \quad ([0, a], [0, b]) \quad \text{or} \quad \left(\left[\frac{a+b}{2}, \frac{a-b}{2} \right], \left[\frac{a+b}{2}, \frac{b-a}{2} \right] \right).$$

Here equivalence is generated by swapping the sequences, cyclically shifting or reversing the first sequence, simultaneous decimation, and simultaneous

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alternating negation, as in [1, §2.1 and Conjecture 1]. They established the conjecture for $v < 100$ and identified $v = 100$ as the first unresolved length [1, §7]. These generators induce the corresponding operations on the length-two compression: a shift or reversal exchanges the parity classes, an admissible decimation is odd and preserves them, and alternating negation changes the sign of the odd class. Negation of one sequence is itself generated by the five operations [1, §2.1].

Theorem 1. *For every integer $n \geq 2$, there is a binary Golay pair of length 10^n whose $(10^n/2)$ -compression is equivalent to neither of the two pairs in (1), with a, b equal to its row sums. Consequently, Conjecture 1 of Lumsden, Kotsireas, and Bright is false in infinitely many lengths. The case $n = 2$ is a least-length counterexample.*

Let $\mathcal{U} = \{\pm 1, \pm i\}$. For a length-two integer pair $H = ([x_0, x_1], [y_0, y_1])$, define

$$z(H) = (x_0 + x_1) + i(y_0 + y_1), \quad w(H) = (x_0 - x_1) + i(y_0 - y_1).$$

Lemma 2. *If H is equivalent to either pair in (1), then*

$$(2) \quad w(H) \in \mathcal{U} z(H) \cup \mathcal{U} \overline{z(H)}.$$

Proof. For the two pairs in (1), respectively, one has $w = -z$ and $w = i\bar{z}$. At length two, swapping the sequences sends

$$(z, w) \mapsto (i\bar{z}, i\bar{w}),$$

shifting or reversing the first sequence sends $(z, w) \mapsto (z, -\bar{w})$, decimation is trivial, and simultaneous alternating negation sends $(z, w) \mapsto (w, z)$. Since $\mathcal{U}$ is closed under multiplication, inversion, and conjugation, a direct substitution shows that each operation preserves (2). $\square$

2. CONSTRUCTION

Proof of Theorem 1. Identify the symbols $+$ and $-$ with 1 and -1 . Consider the length-10 pairs

$$\begin{aligned} U &= +-+++++-, & V &= +---+---+, \\ C &= -+++--+-, & D &= -++++-+-. \end{aligned}$$

With

$$c = (3, 0, 1, 0, -1, -2, 1, 2, -1),$$

direct calculation gives

$$(N_U(s))_{s=1}^9 = (N_D(s))_{s=1}^9 = c, \quad (N_V(s))_{s=1}^9 = (N_C(s))_{s=1}^9 = -c.$$

Thus (U, V) and (C, D) are Golay pairs.

Put

$$p_j = \frac{u_j + v_j}{2}, \quad q_j = \frac{u_j - v_j}{2}.$$

Explicitly,

$$\begin{aligned} p &= (1, -1, -1, 1, 0, 1, 0, 0, 0, 0), \\ q &= (0, 0, 0, 0, 1, 0, 1, 1, 1, -1). \end{aligned}$$

Hence $p_j q_j = 0$, $p_j^2 + q_j^2 = 1$ for every j , and

$$(3) \quad \sum_{j=0}^9 p_j = 1, \quad \sum_{j=0}^9 q_j = 3.$$

For an even-length pair (X, Y) , let X^R denote reversal and define

$$(4) \quad \mathcal{T}(X, Y) = \left(\parallel_{j=0}^9 (p_j X + q_j Y^R), \parallel_{j=0}^9 (-q_j X^R + p_j Y) \right),$$

where $\parallel$ denotes concatenation. This is a standard form of Turyn multiplication [2].

Let $F_X(z) = \sum_j x_j z^j$, and let P, Q be the polynomials with coefficient sequences p, q . If $(A, B) = \mathcal{T}(X, Y)$ and m is the common length of X, Y , then, using

$$F_{X^R}(z) = z^{m-1} F_X(z^{-1}),$$

cancellation of the cross terms gives

$$\begin{aligned} &F_A(z)F_A(z^{-1}) + F_B(z)F_B(z^{-1}) \\ &= (P(z^m)P(z^{-m}) + Q(z^m)Q(z^{-m})) \\ &\quad \times (F_X(z)F_X(z^{-1}) + F_Y(z)F_Y(z^{-1})). \end{aligned}$$

Since $F_U = P + Q$ and $F_V = P - Q$, the Golay identity for (U, V) yields

$$2(P(z)P(z^{-1}) + Q(z)Q(z^{-1})) = 20.$$

Thus the first factor above is identically 10. If (X, Y) is a Golay pair of length m , the second factor is $2m$; therefore (4) is a Golay pair of length $10m$. It is binary because $p_j q_j = 0$ and $p_j^2 + q_j^2 = 1$.

Set $(A_1, B_1) = (C, D)$ and recursively define

$$(A_{n+1}, B_{n+1}) = \mathcal{T}(A_n, B_n).$$

Then (A_n, B_n) is a binary Golay pair of length 10^n . Define

$$z_n = r(A_n) + ir(B_n), \quad w_n = e(A_n) + ie(B_n).$$

If H_n is its $(10^n/2)$ -compression, then $z(H_n) = z_n$ and $w(H_n) = w_n$.

Since the block length in (4) is even, alternating sums add across the concatenation; moreover, $e(X^R) = -e(X)$ for every even-length sequence X . Using (3) in (4) therefore gives

$$\begin{aligned} r(A_{n+1}) &= r(A_n) + 3r(B_n), & r(B_{n+1}) &= -3r(A_n) + r(B_n), \\ e(A_{n+1}) &= e(A_n) - 3e(B_n), & e(B_{n+1}) &= 3e(A_n) + e(B_n). \end{aligned}$$

Consequently,

$$(5) \quad z_{n+1} = (1 - 3i)z_n, \quad w_{n+1} = (1 + 3i)w_n.$$

The initial values are

$$z_1 = 2 + 4i, \quad w_1 = 4 - 2i = -iz_1.$$

Writing $t = 1 + 3i$, equations (5) give

$$(6) \quad z_n = \bar{t}^{n-1} z_1, \quad w_n = -it^{n-1} z_1.$$

Suppose $n \geq 2$ and (2) holds. If $w_n = uz_n$ for some $u \in \mathcal{U}$, then (6) implies that t^{n-1} and $\bar{t}^{n-1}$ are associates in $\mathbb{Z}[i]$. This is impossible, since

$$t = (1 + i)(2 + i), \quad \bar{t} = (1 - i)(2 - i),$$

and $2 + i$ and $2 - i$ are nonassociate Gaussian primes, both of norm 5. Unique factorization in $\mathbb{Z}[i]$ rules out the association.

If instead $w_n = u\bar{z}_n$, cancellation in (6) gives $-iz_1 = u\bar{z}_1$, but

$$\frac{-iz_1}{\bar{z}_1} = \frac{4 + 3i}{5} \notin \mathcal{U}.$$

Thus (2) fails, and Lemma 2 proves the asserted inequivalence.

For $n = 2$, equations (6) give

$$z_2 = 14 - 2i, \quad w_2 = 10 + 10i,$$

so the compression of (A_2, B_2) is

$$H_2 = ([12, 2], [4, -6]).$$

Replacing B_2 by $-B_2$ gives another Golay pair whose compression is

$$\tilde{H}_2 = ([12, 2], [-4, 6])$$

with row sums $a = 14$ and $b = 2$. The two representatives in (1) are then

$$([0, 14], [0, 2]) \quad \text{and} \quad ([8, 6], [8, -6]).$$

The maximum absolute entry is respectively 12, 14, and 8; this quantity is invariant under the length-two equivalence operations, giving a direct check at length 100. Since the conjecture holds for $v < 100$ by [1], this counterexample has least possible length. $\square$

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